

SMART HYDRATION REMINDER PCB

Hardware Design Documentation

Board: 4-layer PCB, 37.48 mm x 29.99 mm

MCU: Espressif ESP32-C3-MINI-1U-H4

IMU: Bosch BMI270

Power: USB-C charging, 1S Li-ion/LiPo cell, 3.3 V LDO with hard-off slide switch

1. DESIGN OVERVIEW

The Smart Hydration Reminder is a compact, low-power, wireless-capable PCB designed to live in a water bottle base or sleeve. It detects when the user drinks using a 6-axis IMU, keeps a running record of intake events, and reminds the user with a buzzer when too much time has passed without drinking. A companion phone app can connect over Bluetooth LE to adjust reminder intervals, mute alerts, read battery status and push firmware updates.

The platform combines motion intelligence, multi-modal alerts, low-power telemetry and dual-mode connectivity (Wi-Fi and Bluetooth LE) to monitor drinking patterns, track the interval between sips and deliver adaptive reminders.

The hardware is organised as four layers, each building on the one before:

Charging layer: safe single-cell Li-ion charging from USB-C (J1, USBLC6-2SC6, MCP73831).

Power-gating layer: a user-operated hard-off that shuts down the entire downstream system (S1, RT9080 LDO).

Compute-and-sense layer: application logic, radio, motion detection and battery monitoring (ESP32-C3-MINI-1U, BMI270).

Feedback layer: audible reminders and visual status (buzzer, LEDs, tactile button).

A central design decision is that the slide switch gates the LDO itself, not just the buzzer. Flipping it to OFF removes power from every downstream block (MCU, IMU, LEDs and buzzer driver), giving true zero downstream draw for storage or travel. The charger and cell remain upstream of the switch, so the battery still charges over USB regardless of the switch position.

2. SYSTEM ARCHITECTURE

Power flows from the USB-C connector (J1) through the charger (U4) into the 1S Li-ion/LiPo cell, and from the cell through the switch-controlled RT9080 LDO (IC1) to the 3.3 V rail. The 3.3 V rail supplies the ESP32-C3 (U2), the BMI270 IMU (U3) and the buzzer. USB data lines (D+ and D-) pass through an ESD protection array (U6) to the MCU. The IMU communicates with the MCU over I2C and signals motion through an interrupt line. The battery voltage is sampled by the MCU through a resistor divider. The MCU drives the battery LEDs (D2, D3) and the buzzer driver (Q1), and reads the snooze/log push button (S2).

3. POWER SUBSYSTEM

3.1 USB-C Input and ESD Protection

Connector: USB Type-C 16-pin receptacle (J1), used for USB 2.0 data (D+ and D-) and VBUS/GND.

CC termination: 5.1 kOhm pull-down resistors (R1, R2) on CC1 and CC2 identify the board as a sink, so a Type-C source enables VBUS at 5 V.

Transient protection: a TVS diode array (U6, USBLC6-2SC6) sits directly beside the receptacle and clamps electrostatic discharge on the data lines, rated to +/-15 kV air discharge (IEC 61000-4-2).

3.2 Battery Charging

Charge controller: Microchip MCP73831T-2ACI/OT (U4) in a SOT-23-5 package, using constant-current / constant-voltage charging to 4.2 V.

Charge current: set by resistor R5 on the PROG pin according to $I_{CHG} = 1000 \text{ V} / R_{PROG}$. With R5 = 2 kOhm, the fast-charge current is 500 mA.

Status indication: the open-drain STAT pin drives the charge LED (D1) through R4 = 4.7 kOhm. D1 is lit during the constant-current and constant-voltage phases and turns off (high impedance) once charging terminates.

Cell: a 1S Li-ion/LiPo cell (4.2 V maximum) connects at pads J2 and J3. A cell with a built-in protection circuit is required, as the board has no over-discharge protection of its own.

3.3 Hard-Off Power Switch

Switch: C&K JS202011SCQN (S1), a subminiature surface-mount SPDT slide switch.

Rationale: rather than routing the ESP32-C3's high RF supply current (transmit spikes of around 350 mA) through small mechanical switch contacts, which causes contact wear and voltage droop, the switch drives the high-impedance enable (EN) input of the LDO.

Wiring:

- Pin 1 (outer): VBAT
- Pin 2 (common): EN pin of IC1
- Pin 3 (outer): GND
- R13 = 100 kOhm additionally ties EN to GND.

OFF state: EN is driven to GND, the LDO enters shutdown (typically below 0.1 uA) and nothing downstream draws current.

Charging is independent of the switch. The charger and cell sit upstream of the switch, so a switched-off device still charges.

Trade-off: when switched off, the MCU cannot log the event, cannot save state beyond what is already in flash, and cannot be reached over BLE. This is intentional for a hard off.

Control roles: the tactile button (S2) is the fast, momentary snooze and silence control. The slide switch serves purely as the full power gate.

3.4 3.3 V Regulation

Regulator: Richtek RT9080-33GJ5 (IC1) in a TSOT-23-5 package.

Quiescent current: about 2 μ A typical, compared with roughly 5 mA for a classic AMS1117-class regulator, which would drain a small cell in days.

Output capability: 600 mA continuous, above the ESP32-C3 peak transient demand (about 500 mA system requirement).

Dropout: 310 mV at 600 mA. Dropout scales with load, so at the light average loads of this application the regulator holds 3.3 V until the cell is nearly empty (about 3.0 V). Under a full 600 mA load, regulation needs roughly 3.6 V at the input. The ESP32-C3 brown-out detector should be used for the low-voltage end.

4. COMPUTE AND CONNECTIVITY

4.1 Microcontroller

Module: Espressif ESP32-C3-MINI-1U-H4 (U2).

Core: 32-bit single-core RISC-V, up to 160 MHz, with 4 MB embedded flash.

RF: U.FL connector for an external antenna, which avoids the detuning that liquid volumes inside a water bottle would cause on an on-board antenna.

The MCU runs the whole system: timers, the buzzer state machine, IMU handling, ADC readings, intake logging and the BLE/Wi-Fi link. Choosing a radio-capable MCU means app connectivity costs no additional hardware.

4.2 Mobile Ecosystem and Firmware Interoperability

BLE control: a Bluetooth 5 LE GATT server lets the companion app modify reminder intervals, mute acoustic alerts, read real-time battery status and push over-the-air (OTA) firmware updates.

Intake logging: drinking events are stored locally in non-volatile storage (NVS flash) and batch-synchronised to the cloud when Wi-Fi is reachable.

Radio versus battery life: BLE activity draws tens of milliamps while active, against microamps in deep sleep. The connection and advertising schedule directly trades app responsiveness (for example, muting a reminder remotely) against battery life, and is a key firmware tuning parameter.

5. MOTION SENSING AND DRINKING DETECTION

5.1 Sensor

IC: Bosch BMI270 (U3), a 16-bit ultra-low-power 6-axis IMU combining a 3-axis accelerometer and a 3-axis gyroscope.

Power: below 685 uA in full-performance mode, and a few microamps in suspend.

Interface: I2C on IO2 (SDA) and IO3 (SCL) with 4.7 kOhm pull-ups (R7, R8) and decoupling on C6 and C7. CSB must be strapped high for I2C mode and SDO strapped to set the address.

5.2 Drinking Detection Algorithm

Motion processing runs directly in firmware:

1. Pickup detection. High-pass-filtered Z-axis acceleration thresholds detect the bottle being lifted from a resting surface.
2. Tilt estimation. Pitch and roll dynamics give the tilt angle relative to gravity: $\theta_{\text{tilt}} = \text{atan2}(A_y, \sqrt{A_x^2 + A_z^2})$.
3. Sip validation. A sip event is confirmed when the tilt exceeds 45 degrees for at least 1.5 seconds, followed by a controlled lowering movement.
4. Timer reset. Each verified sip resets the reminder countdown, so users who hydrate naturally are never nagged.

5.3 Deep Sleep with Wake-on-Motion

BMI270 INT1 (pin 4) connects to the RTC-capable pin IO4 of the ESP32-C3. When idle, the ESP32-C3 sits in deep sleep (about 5 uA) while the BMI270 stays in its low-power motion-detect mode. Moving the bottle makes the BMI270 pulse INT1, waking the MCU to update intake telemetry and timer logs.

6. USER INTERFACE AND FEEDBACK

6.1 Battery Voltage Monitoring

Divider: $R14 = R15 = 100 \text{ kOhm}$ halve the battery voltage, so $V_{\text{ADC}} = V_{\text{BAT}} \times R15 / (R14 + R15) = V_{\text{BAT}} / 2$.

Range: at $V_{\text{BAT}} = 4.2 \text{ V}$, $V_{\text{ADC}} = 2.1 \text{ V}$, which is inside the ESP32-C3 ADC range at 11 dB attenuation (0 to about 2.5 V). The divider feeds ADC pin IO0.

Noise filter: C10 = 0.1 uF across R15 forms a low-pass filter with the 50 kOhm Thevenin resistance, giving $f_c = 1 / (2 \times \pi \times 50 \text{ kOhm} \times 0.1 \text{ uF})$, approximately 31.8 Hz. This suppresses ripple from Wi-Fi and Bluetooth transmit bursts.

Firmware note: the ESP32-C3 ADC is non-linear at the ends of its range, so the calibrated ADC API (eFuse calibration values) should be used rather than a simple linear conversion.

6.2 Visual Battery Telemetry

Green LED (D3): driven by IO5 through a 470 Ohm resistor (R11). Lit when V_BAT is 3.5 V or higher.

Red LED (D2): driven by IO6 through a 470 Ohm resistor (R12). Lit when V_BAT is below 3.5 V.

Charge LED (D1): driven by the MCP73831 STAT pin through 4.7 kOhm (R4). Lit while charging.

Firmware drives IO5 high and IO6 low above the threshold, and the reverse below it.

6.3 Acoustic Alert Subsystem

Transducer: TMB12A03 (U5), a 3 V active electromagnetic buzzer.

Driver: Alpha and Omega AO3400 N-channel MOSFET (Q1, SOT-23-3, V_DS = 30 V, I_D = 5.8 A) low-side switches the buzzer.

Gate drive: IO7 (BUZZ net). R6 = 10 kOhm pulls the gate to GND so the buzzer cannot turn on during MCU reset or boot.

Flyback protection: a 1N4148 diode (CR1) across the buzzer terminals clamps inductive spikes at turn-off.

Patterns: the MCU gates IO7 using LEDC PWM to create beep rhythms, such as a gentle single chime versus an escalating pattern after ignored reminders. The same driver circuit serves every pattern.

6.4 Snooze and Log Button

Switch: C&K PTS526 (S2), a low-profile tactile switch.

Circuit: IO1 with a 10 kOhm pull-up (R10) to 3.3 V and a 0.1 uF debounce capacitor (C9) to GND.

Short press (under 1 second): acknowledges and silences an active alarm and resets the reminder timer.

Long press (over 3 seconds): manual logging trigger, or initiates Bluetooth pairing.

7. PCB DESIGN

The board is a four-layer PCB measuring 37.48 mm x 29.99 mm, laid out to be as compact as possible so it fits inside a bottle base or sleeve.

7.1 Layer Stack-Up

Layer 1 (top): power distribution and component placement.
Layer 2: continuous ground plane.
Layer 3: signal routing.
Layer 4 (bottom): power and ground pours and remaining routing.

The dedicated ground plane on layer 2 gives every top-layer signal a solid return path and keeps the RF-adjacent ESP32-C3 section and the mixed analogue/digital sections (ADC divider, I2C sensor) well referenced. Keeping signals on layer 3 between planes reduces coupling and improves noise immunity, and the power and ground pours on layer 4 keep supply impedance low across the board.

7.2 Placement

- USB-C (J1) on the left board edge for easy cable access, with the ESD array and CC resistors immediately beside it.
- ESP32-C3 module (U2) at the top centre, close to the board edge to keep the radio section clear of other circuitry.
- BMI270 (U3) with its pull-ups and decoupling near the MCU, keeping the I2C and interrupt traces short.
- Charger (U4), LDO (IC1) and passives grouped in the lower centre, keeping the power path compact.
- Slide switch (S1) and tactile button (S2) on the right edge for user access.
- Buzzer (U5) and driver (Q1) at the top right.
- Battery pads (J2, J3) on the left side beside the charging circuitry.

(Insert the 3D render of the board here.)

8. POWER CONSIDERATIONS

Standby battery life is set by the sum of the always-connected loads plus the deep-sleep floor of the MCU and IMU. From the values used in this design:

- RT9080 quiescent current: about 2 μA (datasheet typical).
- ESP32-C3 deep sleep: about 5 μA (datasheet typical).
- BMI270 in motion-detect mode: to be measured on hardware, since the accelerometer must stay active.
- EN pull-down R13 with the switch ON: about 42 μA ($4.2\text{ V} / 100\text{ k}\Omega$).
- Battery divider R14 + R15: about 21 μA ($4.2\text{ V} / 200\text{ k}\Omega$).

Radio activity dominates when active, so the BLE connection interval and advertising schedule should be chosen with a measured energy budget in mind.

9. KNOWN LIMITATIONS AND PLANNED IMPROVEMENTS

Always-on resistive loads: R13 and the R14/R15 divider account for most of the standby current. Moving to values of about 1 MOhm or higher, or gating the divider from a GPIO, would substantially extend standby life.

Charge current: 500 mA on a 450 mAh cell is roughly a 1.1C rate. Reducing it to about 250 mA (R5 of about 4 kOhm) is gentler on the cell and limits heat in the linear charger.

Buzzer control: the active buzzer has a fixed internal tone, so PWM gives rhythm control rather than true volume or pitch control. A passive buzzer would enable both.

Battery indication: a single 3.5 V threshold does not map linearly to state of charge. A lookup-table mapping and a blinking red LED for a critical level would improve the indication with no hardware change.

Cell protection: the design assumes a protected cell. On-board over-discharge protection is a candidate for a future revision.

10. CONCLUSION

The Smart Hydration Reminder PCB packs a complete wireless, motion-aware product into a 37.48 mm x 29.99 mm four-layer board. The charging path, LDO-gated hard-off switch and low-quiescent-current regulator form a power architecture built around a small battery, where a switched-off device draws essentially nothing and still charges. The ESP32-C3 and BMI270 combination delivers sip detection, wake-on-motion, local logging and app connectivity on a single radio-capable MCU, while the buzzer and LED feedback keep the user interface simple and robust.

The four-layer stack-up with a dedicated ground plane keeps the compact layout electrically quiet, and the documented limitations give a clear path to a longer-lasting, more refined next revision.

Smart Hydration Reminder | 4-layer hardware platform | 37.48 mm x 29.99 mm